

```
In [2]: import numpy as np
import pandas as pd
from sklearn import svm
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(font_scale=1.2)

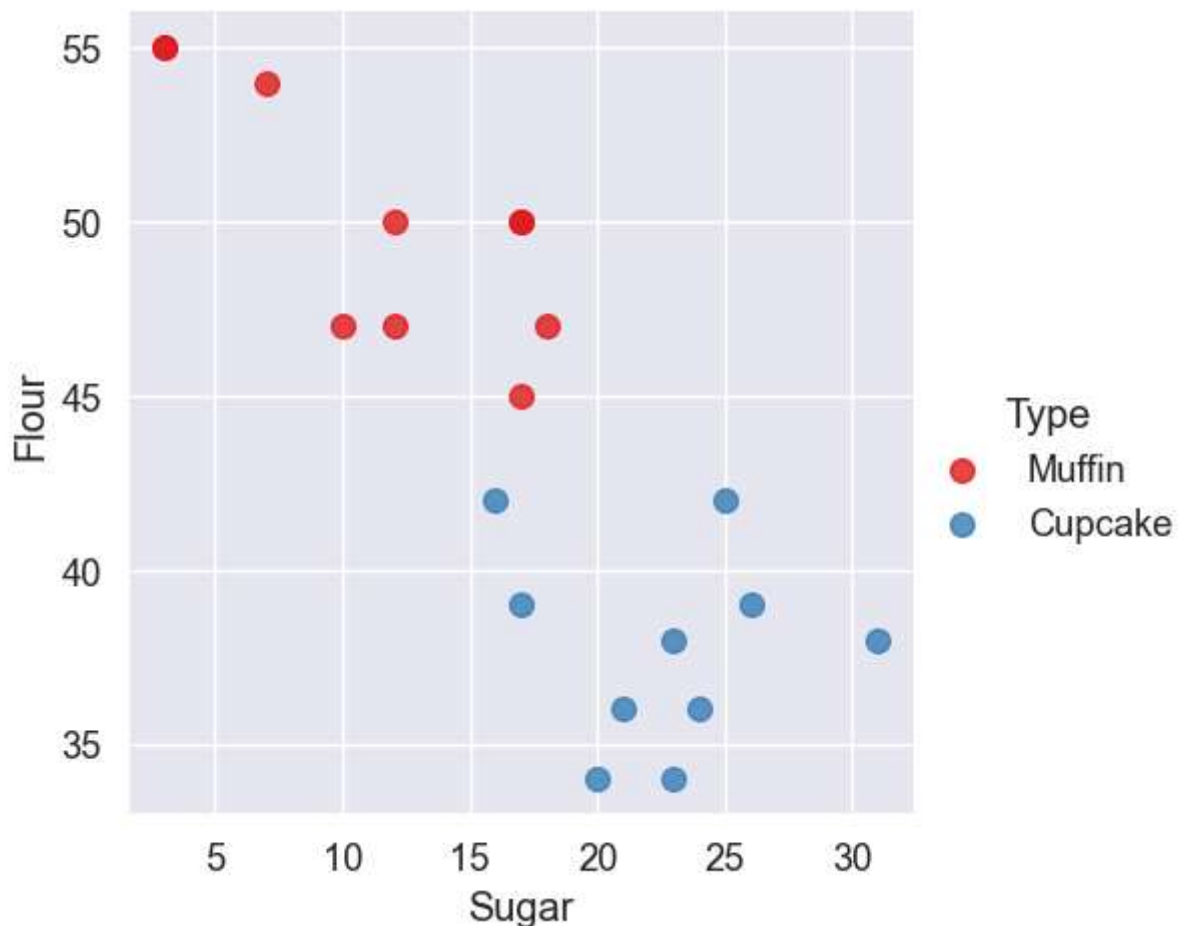
recipes = pd.read_csv('recipes_muffins_cupcakes.csv')
print(recipes.head())
print(recipes.shape)
```

|   | Type   | Flour | Milk | Sugar | Butter | Egg | Baking Powder | Vanilla | Salt |
|---|--------|-------|------|-------|--------|-----|---------------|---------|------|
| 0 | Muffin | 55    | 28   | 3     | 7      | 5   | 2             | 0       | 0    |
| 1 | Muffin | 47    | 24   | 12    | 6      | 9   | 1             | 0       | 0    |
| 2 | Muffin | 47    | 23   | 18    | 6      | 4   | 1             | 0       | 0    |
| 3 | Muffin | 45    | 11   | 17    | 17     | 8   | 1             | 0       | 0    |
| 4 | Muffin | 50    | 25   | 12    | 6      | 5   | 2             | 1       | 0    |

(20, 9)

```
In [3]: sns.lmplot(x='Sugar', y='Flour', data=recipes, hue='Type',
                  palette='Set1', fit_reg=False, scatter_kws={"s":70})
```

Out[3]: <seaborn.axisgrid.FacetGrid at 0x2978ebe7890>



```
In [4]: sugar_butter = recipes[['Sugar', 'Butter']].values
type_label = np.where(recipes['Type']=='Muffin', 0, 1)
```

```
model = svm.SVC(kernel='linear')
model.fit(sugar_butter, type_label)

w = model.coef_[0]
a = -w[0] / w[1]

xx = np.linspace(5,30)
yy = a * xx - (model.intercept_[0] / w[1])

b = model.support_vectors_[0]
yy_down = a * xx + (b[1] - a * b[0])

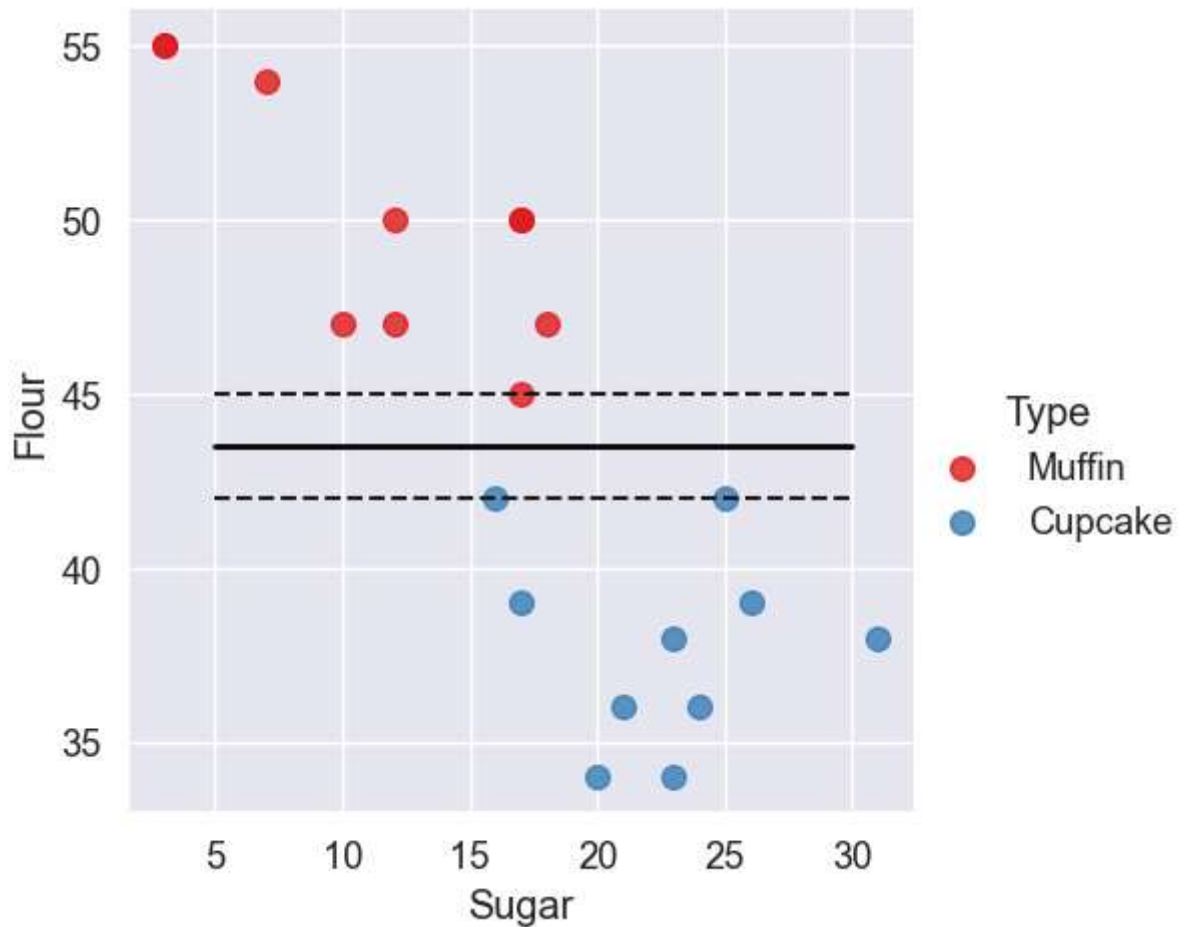
b = model.support_vectors_[-1]
yy_up = a * xx + (b[1] - a * b[0])

sns.lmplot(x='Sugar', y='Flour', data=recipes, hue='Type',
           palette='Set1', fit_reg=False, scatter_kws={"s":70})

plt.plot(xx, yy, linewidth=2, color='black')
plt.plot(xx, yy_down, 'k--')
plt.plot(xx, yy_up, 'k--')

plt.scatter(model.support_vectors_[0],
            model.support_vectors_[1],
            s=80, facecolors='none')

plt.show()
```



```
In [6]: from sklearn.metrics import confusion_matrix, classification_report
        from sklearn.model_selection import train_test_split

        x_train, x_test, y_train, y_test = train_test_split(
            sugar_butter, type_label, test_size=0.2
        )

        model1 = svm.SVC(kernel='linear')
        model1.fit(x_train, y_train)

        pred = model1.predict(x_test)
        print(pred)
```

```
[1 1 0 1]
```

```
In [7]: print(confusion_matrix(y_test, pred))
        print(classification_report(y_test, pred))
```

|                  |              |           |        |          |         |
|------------------|--------------|-----------|--------|----------|---------|
| [[1 0]<br>[0 3]] |              |           |        |          |         |
|                  |              | precision | recall | f1-score | support |
|                  | 0            | 1.00      | 1.00   | 1.00     | 1       |
|                  | 1            | 1.00      | 1.00   | 1.00     | 3       |
|                  | accuracy     |           |        | 1.00     | 4       |
|                  | macro avg    | 1.00      | 1.00   | 1.00     | 4       |
|                  | weighted avg | 1.00      | 1.00   | 1.00     | 4       |